Daniele Della Cioppa

Identity and Access Management Engineer
London, UK

GitHub
LinkedIn

Profile

Italian software engineer based in London, transitioning from product and systems engineering into Identity & Access Management (IAM). I combine hands-on implementation with enterprise identity thinking across Active Directory, Kerberos-based Single Sign-On (SSO), token services and secure backend integration. I am particularly motivated by how trust is established and propagated across operating systems, identity providers and downstream services in Zero Trust architectures.

Work Experience

2025–2026 Current

Software Developer — Akhter Computers (IAM Platform Engineering)

My current role evolved from Android and endpoint development into Identity & Access Management (IAM), where I am designing and implementing a complete identity-driven file access platform from scratch. The core objective is simple and strict: users authenticate once on Windows through Active Directory, and only that verified identity can access protected files and services.

I engineered UserProbe (Windows C#/ .NET desktop client), the Akhter Identity Provider (Python FastAPI), and a token-validated FileServer, supported by Samba Active Directory laboratories for realistic integration and validation. The Identity Provider validates Kerberos/SPNEGO tickets through GSSAPI, issues RS256 JWT tokens, and publishes JWKS endpoints for signature trust distribution. The FileServer validates JWT signatures and derives effective identity through `user_key = iss|sub` for deterministic principal mapping.

To reduce attack surface, I designed secure manual keytab provisioning so critical Kerberos secrets are never exposed over the network. I also package services for production using Debian packages and systemd units, and I maintain extensive LaTeX operational documentation covering installation, deployment and runbooks. This work applies Zero Trust principles end to end, including least privilege, identity propagation, auditable trust boundaries and practical controls around SPNs, service principals, keytabs, RBAC, MFA and Conditional Access patterns (including transferable Microsoft Entra ID / Azure AD knowledge).

2024–2025

Android System Developer — Distributed Device Systems

Built distributed digital-signage and device-synchronisation features with a local-first architecture, improving resilience when cloud connectivity was limited and strengthening endpoint orchestration patterns transferable to identity-aware systems.

2023–2024

Android System Developer — Remote MDM & Update Infrastructure

Developed Android endpoint-control and update infrastructure, including secure APK delivery, remote management workflows, signature/hash verification and controlled command execution across Linux-backed services and mobile clients.

2021–2023

Software Development Apprenticeship

Contributed to IoT and Android projects (telemetry, gateway communication, PostgreSQL-backed monitoring and field diagnostics), building the systems foundation later applied to security-focused identity architecture.